

Assignment 6 - Matrices in R

Exercise 1

```
m1[,2] <- m1[,1] + m1[,3]
```

Exercise 2

1) Create a 6×6 matrix mTest with all 36 elements equal to 0.

```
mTest <- matrix(0, nrow=6, ncol=6)
```

2) Check what the functions row and col return when applied to mTest

```
row(mTest)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	1	1	1	1	1	1
[2,]	2	2	2	2	2	2
[3,]	3	3	3	3	3	3
[4,]	4	4	4	4	4	4
[5,]	5	5	5	5	5	5
[6,]	6	6	6	6	6	6

```
col(mTest)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	1	2	3	4	5	6
[2,]	1	2	3	4	5	6
[3,]	1	2	3	4	5	6
[4,]	1	2	3	4	5	6
[5,]	1	2	3	4	5	6
[6,]	1	2	3	4	5	6

3) Accordingly, create the following matrix m2:

```
m2 <- mTest  
m2[row(mTest) == col(mTest) + 1] <- 1  
m2[row(mTest) == col(mTest) - 1] <- 1
```

Exercise 3

```
m3 <- outer(0:9, 0:9, "+")%%10
```

Exercise 4

```
m4 <- outer(0:8, 0:8, "-")%%9
```